PRANAV GOYAL

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RESEARCH INTERESTS

Social Robot Navigation, Human-Aware Planning, Human-Robot Communication, Planning under Uncertainty.

EDUCATION

University Of Michigan - Ann Arbor *Aug 2024 - Aug 2029*
Ph.D Candidate in Robotics, Advisor: Dr. Christoforos Mavrogiannis *CGPA- 4.12/4.00*

Courses: Math for Robotics, Robotic Systems Laboratory, Computational HRI, Robot Learning for Planning and Control, Information Theory and Statistics, Advanced Computer Vision, Mathematics for Generative Models

BITS PILANI, K.K BIRLA GOA CAMPUS *Aug 2019 - Mar 2023*
Bachelor's in Electronics and Instrumentation Engineering *CGPA-8.58/10*

Courses: Modern Control Systems, Neural Networks and Fuzzy Logic, Foundations of Data Science, Signals and Systems, Non Linear Dynamics and Chaos, Industrial Instrumentation and Control

SKILLS

Languages: Python, C/C++, MATLAB, R, Bash
Frameworks: PyTorch, ROS 1/2, Gazebo, OpenCV, Open3D
Planning & Learning: Optimal and sampling-based control (MPPI, convex MPC), diffusion planning, reinforcement learning, human motion prediction
Perception: 3D reconstruction (SfM, point cloud registration), human detection and tracking
Robot Platforms: Hello Robot Stretch 2, Unitree Go2, Universal Robots arm, WidowX arm, AgileX Ranger Mini 3.0, Motion Capture System (OptiTrack)

PUBLICATIONS

1. **Rethinking Legibility in Social Robot Hallway Navigation: Impact of Intent Representation and Human Distraction.**
Pranav Goyal, *Andrew Stratton, Christoforos Mavrogiannis.*
ACM Transactions on Human-Robot Interaction (THRI), 2026.
2. **Alpaca: An Interactive Platform for Social Robot Navigation.**
Pranav Goyal, *Jiamu Liu, Andrew Stratton, Tisha Jain, Christoforos Mavrogiannis.*
Under review, 2027 ACM/IEEE International Conference on Human-Robot Interaction (HRI).
3. **Rethinking Legibility in Social Robot Navigation: Impact of Intent Representation and Human Distraction.**
Pranav Goyal, *Andrew Stratton, Christoforos Mavrogiannis.*
2026 ACM/IEEE International Conference on Human-Robot Interaction (HRI), Late-Breaking Report.
4. **How Human Motion Prediction Quality Shapes Social Robot Navigation Performance in Constrained Spaces.**
Andrew Stratton, Prudhvi Teja Singamaneni, **Pranav Goyal**, *Rachid Alami, Christoforos Mavrogiannis.*
2026 ACM/IEEE International Conference on Human-Robot Interaction (HRI).
5. **Bi3: A Biplatform, Bicultural, Biperson Dataset for Social Robot Navigation.**
Andrew Stratton, Prudhvi Teja Singamaneni, **Pranav Goyal**, *Rachid Alami, Christoforos Mavrogiannis.*
2026 IEEE International Conference on Robotics and Automation (ICRA).
6. **SocRATES: Towards Automated Scenario-based Testing of Social Navigation Algorithms.**
Shashank Rao Marpally, **Pranav Goyal**, *Harold Soh.*
IEEE Robotics and Automation Letters (RA-L), 2026.

7. **ACME: A Multi-Cultural, Multi-Embodiment Social-Navigation Dataset.**
Shashank Rao Marpally, . . . , **Pranav Goyal**, *. . . , Christoforos Mavrogiannis, Harold Soh.*
Under Review, International Journal of Robotics Research (IJRR).

8. **LTLDoG: Satisfying Temporally-Extended Symbolic Constraints for Safe Diffusion-based Planning.**
Zeyu Feng, Hao Luan, **Pranav Goyal**, *Harold Soh.*
IEEE Robotics and Automation Letters (RA-L), 2024.

9. **Towards Automated Void Detection for Search and Rescue with 3D Perception.**
Ananya Bal, **Pranav Goyal**, *Ashutosh Gupta, David Merrick, Robin Murphy, Howie Choset.*
2023 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS).

10. **A Comparison of Point Cloud Registration Techniques for On-Site Disaster Data from the Surfside Structural Collapse.**
Ananya Bal, Robert Ladig, **Pranav Goyal**, *John Galeotti, Howie Choset, David Merrick, Robin Murphy.*
2022 IEEE International Symposium on Safety, Security, and Rescue Robotics (SSRR).

11. **Towards Automated Scenario Testing of Social Navigation Algorithms.**
Shashank Rao Marpally, **Pranav Goyal**, *Harold Soh.*
Workshop on Unsolved Problems in Social Robot Navigation, Robotics: Science and Systems (RSS), 2024. *Runner-up Best Paper Award.*

RESEARCH EXPERIENCE

Fluent Robotics Lab, University of Michigan - Ann Arbor — Ann Arbor, USA
Graduate Research Assistant, Advisor: Dr. Christoforos Mavrogiannis — *Aug 2024 - Present*

- Developed MPPI-based goal-intent expression algorithms in PyTorch for human-aware navigation, and ran a study on the Hello Robot Stretch 2 platform showing how contextual factors such as human distraction change what makes robot motion legible (HRI 2026 LBR; THRI 2026).
- Building Bi3 and ACME, scenario-centric social navigation datasets spanning two robot platforms and two cultural contexts, for training and benchmarking foundational navigation models (ICRA 2026; under review at IJRR).
- Quantified how the quality of human motion prediction propagates into downstream social navigation performance in constrained spaces, across multiple planners and prediction models (HRI 2026).
- Formalizing planning under actionable uncertainty arising from context such as scene semantics, and generating robot motion that acts on it by being simultaneously legible and information-gathering.
- Developing Alpaca, an open hardware and ROS 2 platform integrating pedestrian tracking, human motion prediction, navigation, and conversational interaction for reproducible field benchmarking (under review at HRI 2027).

CLeAR Lab, National University of Singapore — Singapore
Research Assistant, Advisor: Dr. Harold Soh — *Aug 2023 - Aug 2024*

- Formulated LTLDoG, an LTL-satisfaction-based diffusion global planning framework for navigation in context-rich environments, and deployed it on the Unitree Go2 quadruped (RA-L 2024).
- Designed SocRATES, an LLM-powered scenario generation tool that turns natural-language situation descriptions into executable ROS2/Gazebo benchmarks for social navigation algorithms (RA-L 2026; Runner-up Best Paper, RSS 2024 workshop).
- Studied navigation strategies for quadrupedal robots in dense, unstructured urban environments such as university canteens, including full-stack deployment and data collection on hardware.

Biorobotics Lab, Robotics Institute, CMU — Pittsburgh, USA
Research Intern, Advisor: Dr. Howie Choset — *Jun 2022 - Jan 2023*

- **Project VeRT (DARPA), Bachelor's thesis:** Implemented convex MPC and kinematics-based geometric control for a legged-wheeled robot, and built a working hardware prototype from HEBI modules. [video]
- **NSF Surfside:** Devised an automated void detection pipeline for search-and-rescue, built on point cloud registration and Structure-from-Motion reconstruction of drone footage of the Surfside collapse site (IROS 2023).

- **On-Orbit Manipulation (Northrop Grumman):** Ran hardware-in-the-loop validation of MRV-MEP insertion control on a Universal Robots arm, and developed impedance matching and contact parameter estimation.

Mercedes-Benz Research and Development India Bangalore, India
Computer Vision / Deep Learning Intern *Mar 2023 - May 2023*

- Led a project on trajectory planning for 3D face scanning using a WidowX robot arm, covering both the scanning motion design and the reconstruction pipeline.
- Developed an ArUco-marker-based system for ground-truth head position (Ho/Hp) detection of seated occupants, used to validate in-cabin perception models.

HONORS & AWARDS

- **Robotics PhD Departmental Fellowship:** Awarded first-year fellowship for the PhD studies by the Department of Robotics, University of Michigan, Ann Arbor, MI.
- **Rackham Graduate Student Travel Grant:** Awarded by the Rackham Graduate School, University of Michigan, to present research at an international conference.
- **Runner-up Best Paper Award:** Workshop on Unsolved Problems in Social Robot Navigation, Robotics: Science and Systems (RSS), 2024.

TEACHING AND VOLUNTEERING EXPERIENCE

Graduate Student Instructor, ROB 204: Introduction to Human-Robot Systems *Fall 2026*
Department of Robotics, University of Michigan
Course Developer, ROB 498/599: Computational HRI *Jul 2025 - Oct 2025*
Developed course assignments for the Department of Robotics, University of Michigan
Teaching Assistant, Software for Embedded Systems *Jan 2022 - Jun 2022*
The course is part of CS Master program at Bits Goa under the guidance of Dr. Neena Goveas
Teaching Head, Electronics and Robotics Club *Aug 2021 - Apr 2022*
Organized courses and competitions for inculcating robotics culture in college

PROFESSIONAL SERVICE & MENTORSHIP

- **Review Service:** THRI, RA-L, ICRA, IROS, HRI, Autonomous Robots, RO-MAN.
- **Mentoring:** Jiamu Liu (UMich Robotics M.S.), Atharva Ghotavadekar (BITS Pilani Goa B.E.; now Research Assistant, CLeAR Lab, National University of Singapore).